

Question_4:

Code:

```
1  /*
2  Question_4: Count X in [L,R] for which gcd(X, hex-digit-sum(X)) > 1.
3  */
4
5  #include <iostream>
6  using namespace std;
7
8  int hexSum(int n) {
9      int s = 0;
10     while (n) {
11         s += n % 16;
12         n /= 16;
13     }
14     return s;
15 }
16 int gcd(int a, int b) {
17     while (b) {
18         int t = a % b;
19         a = b;
20         b = t;
21     }
22     return a;
23 }
24 int main() {
25     int T;
26     cin >> T;
27
28     while (T--) {
29         int L, R, ans = 0;
30         cin >> L >> R;
31         for (int x = L; x <= R; x++)
32             if (gcd(x, hexSum(x)) > 1)
33                 ans++;
34         cout << ans << '\n';
35     }
36     return 0;
37 }
```

Output:

Test Case_1:

```
3
1 3
2
5 8
4
7 12
6
PS C:\Users\Prabin Yadav\D
```

Test Case_2:

```
10
15 64
29
14 100
52
9 79
45
19 63
26
3 83
53
5 61
37
18 93
45
20 98
47
13 71
36
12 53
26
PS C:\Users\Prabin Yadav\
```